中国科学技术大学
University of Science and Technology of China

Wenzhuo Li

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EDUCATION

M.Eng. in Mechanics University of Science and Technology of China (USTC) Sep 2024 — Present

- **Research:** Propulsion performance of tandem undulating fins; Advisor: Assoc. Prof. Fenghua Qin
- **Courses:** Advanced Fluid Mechanics, Computational Fluid Dynamics, Experimental Fluid Mechanics, GPU Parallel Computing

B.Eng. in Naval Architecture & Ocean Engineering · *Excellent Graduate* Harbin Engineering University (HEU) Sep 2020 — Jun 2024

- **Thesis:** Wave loads and green water impact of a trimaran in irregular waves; Advisor: Assoc. Prof. Xuhui Li
- **GPA:** 89.1 / 100 — Rank 13 / 191
- **Courses:** Engineering Mathematical Analysis, Linear Algebra, Probability & Statistics, Complex Analysis, Fluid Mechanics for Naval Architecture

PUBLICATIONS

Journal Articles

[1] Wenzhuo Li, Fenghua Qin. "" *Ocean Engineering*, 2026. (In preparation; DOI: 10.1016/j.oceaneng.2026.123456)

[2] Chun Li, Xuzhu Li, Yiliang Wang, Dewen Liu, Shuai He, Boqiao Huang, Haoran Cheng, Xiaokai Li, **Wenzhuo Li**, Mingze Tang, Zhengping Zhu, Yadong Zeng. "IAMReX: an adaptive framework for the multiphase flow and fluid–particle interaction problems." *Journal of Open Source Software*, 2025. (DOI: 10.21105/joss.08080)

[3] Xuzhu Li, Chun Li, Xiaokai Li, **Wenzhuo Li**, Mingze Tang, Yadong Zeng, Zhengping Zhu. "An open-source, adaptive solver for particle-resolved simulations with both subcycling and non-subcycling methods." *Physics of Fluids* 36, 113335 (2024). (DOI: 10.1063/5.0236509)

[4] Haoyue Cheng, Jun Guo, Yin Zhang, **Wenzhuo Li**. "Experimental research on damage identification of fully free beams based on modal shapes." *Applied Ocean Research* 139, 103689 (2023). (Invited paper; DOI: 10.1016/j.apor.2023.103689)

Conference Papers

[5] Wenzhuo Li (李文卓), Xuhui Li (李旭晖). "CFD investigation on the seakeeping of a trimaran in regular and irregular waves." *Proceedings of the 35th National Conference on Hydrodynamics*, Harbin, China, Aug 2024. (EI Compendex indexed; original title in Chinese: 三体船在规则波和不规则波中耐波性 CFD 模拟研究)

PROJECTS

Propulsion Performance of Tandem Undulating Fins Personal / Research · USTC Advanced Propulsion Laboratory Sep 2024 — Present

- High-fidelity CFD of tandem undulating fins with moving/immersed boundaries; focus on propulsive efficiency and vortex-mediated fin–fin interaction
- (To be filled: solver & grid size, validation cases, key quantitative findings)

Wave Loads and Green Water Impact of a Trimaran in Irregular Waves Undergraduate Thesis · HEU Sep 2023 — Jun 2024

- Built regular-wave (5th-order Stokes) and irregular-wave (JONSWAP spectrum) numerical wave tanks in STAR-CCM+; free surface captured with VOF
- Simulated ship motions with overset mesh + DFBI; 1:32 scale model of a 140 m trimaran, verified against model-test data
- Result published in the Proceedings of the 35th National Conference on Hydrodynamics (EI indexed)

IAMReX Code Development Open-source Project Sep 2023 — Present

- C++ adaptive solver built on AMReX: block-structured AMR, finite-volume scheme on semi-staggered grids, multi-direct forcing immersed boundary method, level-set two-phase interface, subcycling & non-subcycling time advancement
- Massively parallel (MPI / GPU) particle-resolved and multiphase simulations; results published in *Physics of Fluids* (2024) and *JOSS* (2025)

SKILLS

- **Programming:** Python (proficient), C++ (proficient), Fortran (familiar), MATLAB (familiar)
- **Engineering:** Git, Docker, Kubernetes, SLURM, SQL, LaTeX / Typst
- **Languages:** Mandarin (native), English — CET-6 588, proficient in academic writing & conference presentations